

Determination of precise harmonic force constants for alanine polypeptides

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Received 13 March 2007; received in revised form 5 May 2007; accepted 7 May 2007

Available online 18 May 2007

Abstract

We have developed a precise harmonic force field for alanine polypeptides extracted directly from the Hessian tensor of density functional theory calculations. The results yield force constants with very small deviations among the bonds and angles of different size of polypeptides providing a force field that reproduces the vibrational spectrum needed to perform molecular dynamics simulations focusing in the use of the vibrational spectrum to transfer information. In this first principles method, no fitting procedures are used in the formalism for the generation of force field constants.

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Keywords: Harmonic approximation; Alanine polypeptides; DFT; FUERZA

1. Introduction

The existing data bases for alanine polypeptides were developed using experimental information and in other cases using Hartree–Fock calculations. Earlier, several force fields were adopted for related molecules, in many cases they were obtained from ab initio calculations and experiment as done for instance by Pulay et al. [1]. Theoretical and experimental force fields were obtained later on for linear polyenes by Zerbetto et al. [2] and by Gussoni et al. [3]. Later on, the advent of the universal force fields (UFF) [4] with the charge equilibration method [5] by Rappe et al. improved tremendously the calculations of several organic molecules as well as some inorganic ones.

With the advent of molecular electronics and nanotechnology the need for very precise force fields for molecular mechanics (MM) and molecular dynamics (MD) simulations is very important for a precise description of atomis-

tic features of large systems that could be prohibitive to deal with ab initio methods, even when density functional theory (DFT) based methods are used.

In this work, we use the program FUERZA [6] to implement harmonic force field parameters for alanine polypeptides. In the next section we provide a short summary of the theory and in the third section we provide results and discussions.

2. Procedure and methodology

First we optimized the geometries using the B3PW91 hybrid functional, which uses a combination of the Becke-3 (B3) [7] exchange functionals and the Perdew-Wang (PW91) [8,9] correlation functionals. All the DFT calculations are performed with the 6-31G* [10] basis sets using the program Gaussian-03 [11]. Then we use the program FUERZA, which is a practical procedure to obtain internal force constants from Cartesian second derivatives (Hessians). It allows a systematic analysis of pair atomic interactions in a molecular system given defined as $\delta F_A = [k_{AB}]\delta r_B$ for any two atoms A and B.

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$$\begin{bmatrix} \delta F_{x_A} \\ \delta F_{y_A} \\ \delta F_{z_A} \end{bmatrix} = - \begin{bmatrix} \frac{\partial^2 E}{\partial x_A \partial x_B} & \frac{\partial^2 E}{\partial x_A \partial y_B} & \frac{\partial^2 E}{\partial x_A \partial z_B} \\ \frac{\partial^2 E}{\partial y_A \partial x_B} & \frac{\partial^2 E}{\partial y_A \partial y_B} & \frac{\partial^2 E}{\partial y_A \partial z_B} \\ \frac{\partial^2 E}{\partial z_A \partial x_B} & \frac{\partial^2 E}{\partial z_A \partial y_B} & \frac{\partial^2 E}{\partial z_A \partial z_B} \end{bmatrix} \times \begin{bmatrix} \delta x_B \\ \delta y_B \\ \delta z_B \end{bmatrix}$$

This procedure is invariant to the choice of internal coordinates of the molecule. Thus force constants, k_{AB} , for bonds or for any pair of atoms, A and B, are defined by means of the eigenanalysis of their pair interaction matrix as

$$k_{AB} = \sum_{i=1}^3 \lambda_i^{AB} |\hat{u}^{AB} \times \hat{v}_i^{AB}|$$

where λ_i^{AB} are the eigenvalues of the 3×3 pair interaction matrix, $\hat{u}^{AB}$ is the unit vector between atoms A and B, and $\hat{v}_i^{AB}$ are the eigenvectors of the pair interaction matrix. The angle bending force constant, k_θ is obtained from their corresponding two-pair interaction matrices of the two bonds or distances, AB and BC defining the angle θ , or ABC [6].

$$\frac{1}{k_\theta} = \frac{1}{R_{AB}^2 \sum_{i=1}^3 \lambda_i^{AB} |\hat{u}^{PA} \cdot \hat{v}_i^{AB}|} + \frac{1}{R_{CB}^2 \sum_{i=1}^3 \lambda_i^{CB} |\hat{u}^{PC} \cdot \hat{v}_i^{CB}|}$$

where

$$\begin{aligned} \hat{u}_N &= \frac{\hat{u}^{CB} \times \hat{u}^{AB}}{|\hat{u}^{CB} \times \hat{u}^{AB}|}, \\ \hat{u}^{PA} &= \hat{u}_N \times \hat{u}^{AB}, \quad \text{and} \\ \hat{u}^{PC} &= \hat{u}^{CN} \times \hat{u}_N \end{aligned}$$

The force constant is obtained from an average of the corresponding values obtained from the heptamer, octamer and nonamer polypeptides.

3. Results

Fig. 1 defines the atom types used for the force field of alanine polypeptides according to the results shown in Tables 1 and 2.

Table 1 shows the parameters for the alanine bonds. In all cases the standard deviations for the force constants are relatively small compared to the values of the force constants. A similar outcome is also obtained for the bond lengths. Given the quality of the results, we can state that this represents a very precise harmonic force field for chemical bonds.

Similarly, Table 2 shows the angular parameters for the harmonic force field of the alanine peptides. The standard deviations show a high degree of confidence on the angular force field parameters.

Table 3 compares the harmonic bond force field parameter generated from the FUERZA program with those from the literature. We find out for instance that the AMBER force fields [12] are always smaller than those obtained by our procedure; however, the bond lengths are within a good range of tolerance. The force constant tabulated in the CRC Handbook [13] are larger than those from our procedure except for the one for the double bonded CO. The three bases agree in having the double bonded CO with the highest force constant; however the AMBER basis yields a value underestimated by almost 50%. We can notice that there are some differences even in the trends of the specialized FUERZA force field and the general force fields. For instance, the lowest force constant corresponds to the C1–C in the FUERZA but to C1–C1 in the AMBER and C1–H1 in the CRC, which is a collection of force constant from several sources. However, all three bases agree that the double bonded CO corresponds to the largest force constant. The difference between force field does not mean a deficiency of the force

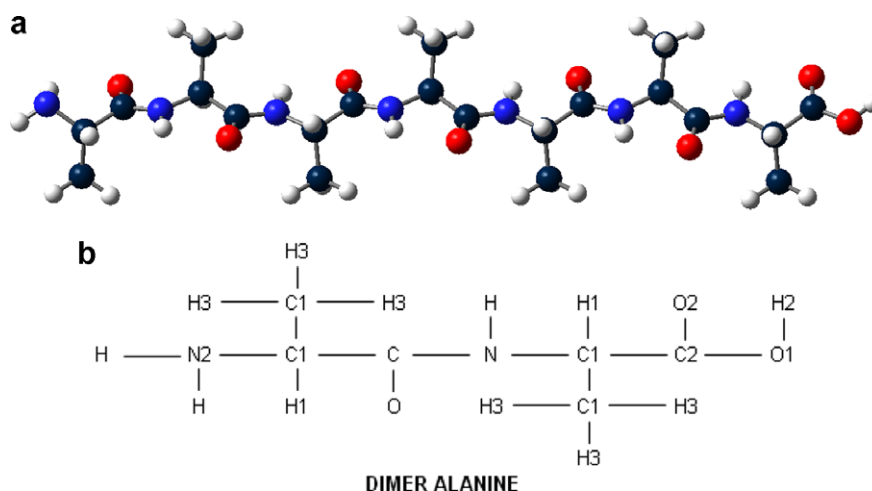


Fig. 1. Examples of molecules studied in this paper, (a) alanine polypeptide (heptamer), atoms for the molecule are color coded, O (red), N (blue), C (black), H (white). (b) Definition of atoms types used in this work. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1

Force constants (k) and bond lengths (b_o) for the alanine polypeptides bonds including their standard deviations and the number (n) of samples used for the average

Bond class	Bond types	k (kcal/mol Å ²)	k Standard deviation (kcal/mol Å ²)	b_o (Å)	b_o Standard deviation (Å)	n
N—H	N—H	1129.0	8.6	1.016	0.001	23
	N2—H	1155.3	0.7	1.013	0.000	3
C—H	C1—H3	938.8	8.6	1.095	0.001	72
	C1—H1	900.6	4.4	1.098	0.001	23
C—N	C1—N	890.3	3.9	1.447	0.001	21
	C1—N2	865.3	3.7	1.457	0.001	3
	C—N	1270.0	12.4	1.348	0.002	21
C—O	C—O	1920.3	8.0	1.231	0.001	21
	C2—O2	2140.4	0.6	1.212	0.000	3
	C2—O1	1148.2	0.1	1.341	0.000	3
C—C	C—C1	734.3	4.2	1.535	0.001	21
	C1—C1	746.1	3.1	1.535	0.001	23
	C1—C2	779.6	1.5	1.517	0.000	3
O—H	O1—H2	1213.0	0.1	0.974	0.000	2

Table 2

Force constants for the alanine polypeptide bending angles including their standard deviations and the number of samples used for the average

Angle class	Angle type	Force constant (kcal/mol deg ²)	Standard deviation (kcal/mol deg ²)	Bending angle (°)	Standard deviation (°)	n
C—C—H	C—C1—H1	76.1	1.5	109.40	1.32	19
	C2—C1—H1	74.0	0.0	108.40	0.00	3
	C1—C1—H3	72.9	2.8	110.89	0.55	36
	C1—C1—H1	52.5	2.5	108.67	0.23	22
C—C—O	C1—C—O	134.8	0.6	121.24	0.13	18
	C1—C2—O2	124.9	0.3	124.54	0.03	3
	C1—C2—O1	156.8	0.4	111.90	0.00	3
O—C—O	O2—C2—O1	146.8	0.1	123.55	0.03	3
C—O—H	C2—O1—H2	71.9	0.0	106.74	0.00	3
C—N—H	C—N—H	73.5	0.7	122.24	0.34	20
	C1—N—H	69.3	1.0	114.47	0.53	21
	C1—N2—H	72.2	2.3	108.31	1.53	5
C—C—N	C1—C1—N	129.3	2.7	112.09	0.04	14
	C1—C1—N2	156.4	0.5	114.65	0.00	3
	C1—C—N	170.9	0.8	115.48	0.06	18
	C—C1—N	173.0	1.1	106.07	0.19	18
	C—C1—N2	147.2	0.1	107.03	0.00	3
	C2—C1—N	143.5	1.1	106.91	0.00	3
N—C—H	N—C1—H1	80.7	0.9	108.60	0.11	20
	N2—C1—H1	83.3	0.9	110.16	2.35	3
N—C—O	N—C—O	131.6	0.3	123.22	0.24	21
C—C—C	C1—C1—C	157.7	3.0	111.46	1.28	20
	C1—C1—C2	134.3	1.7	111.36	0.03	3
C—N—C	C1—N—C	167.0	0.9	122.75	0.32	20
H—C—H	H3—C1—H3	45.3	5.4	108.55	0.40	58
H—N—H	H—N2—H	46.3	0.5	104.51	7.19	3

field but simply means that they were optimized to reproduce different properties.

For the harmonic angular bending parameters, the situation is a little bit different (Table 4). The comparison is more erratic for the bond angle force constants (in kcal/mol deg²) than for the bond lengths. The smaller one

predicted by the FUERZA program is the H3—C1—H3 with a value of 45.3; however the AMBER yields the C1—N—H with 70 as the lowest; however both bases agree that the highest should be C—C1—N angle with 160. The AMBER force field sets the N—C—O, C1—C—O, and O2—C2—O1 also at the highest value of 160 but the

Table 3
Comparison between force constants k (kcal/mol Å²) and bond lengths b_o (Å)

	FUERZA (k_n)	AMBER (k_n)	CRC (k_n)	FUERZA (b_o)	AMBER (b_o)
C—N	1270.0	980	737.8	1.348	1.335
C1—N	890.3	674		1.447	1.449
C1—C	734.3	634	743.5	1.535	1.522
C1—C1	746.1	620		1.535	1.526
C—O	1920.3	1140	2305.5	1.231	1.229
C2—O1	1148.2	900	781.0	1.341	1.364
C1—H1	900.6	680	645.5	1.098	1.09
C1—H3	938.8	680		1.095	1.09
N—H	1129.0	868	860.2	1.016	1.01
O1—H2	1213.0	1106	—	0.974	0.96

Table 4
Comparison of force constant obtained with the force program with those from other sources

	FUERZA (k_θ)	AMBER (k_θ)	FUERZA (θ)	AMBER (θ)
C1—N—C	167.0	100.0	122.75	121.90
C1—C—N	170.9	140.0	115.48	116.60
C—C1—N	173.0	160.0	106.07	109.70
C1—C1—N	129.3	126.0	112.09	110.10
C1—N—H	69.3	60.0	114.47	118.04
C—N—H	73.5	60.0	122.24	120.00
C1—C1—C	157.7	126.0	111.46	111.10
C1—C1—H1	52.5	100.0	108.67	109.50
C—C1—H1	76.1	100.0	109.40	109.50
C1—C1—H3	72.9	100.0	110.89	109.50
N—C1—H1	80.7	100.0	108.60	109.50
C1—C2—O1	156.8		111.90	
C1—C—O	134.8	160.0	121.24	120.40
N—C—O	131.6	160.0	123.22	122.90
H—N2—H	46.3	70.0	104.51	120.00
O2—C2—O1	146.8	160.0	123.55	126.00
C2—O1—H2	71.9	70.0	106.74	113.00
H3—C1—H3	45.3	70.0	108.55	109.50

Angles θ (°) and angular force constant k_θ (kcal/mol deg²).

FUERZA program set them at 132, 135, and 147, respectively. The prediction of the angles is practically similar between the two bases with the largest differences within 3° except for the H—N2—H in which N2 is set as a sp² perhaps by a HF optimization. We get a sp³ character for the N2.

To test if our anzatz to reduce a tensor quantity of rank-2 to a scalar quantity that can be used in standard force fields regardless of the location and direction of the equivalent bonds, we calculate the force fields for all equivalent bonds in a cyclic molecule such as benzene. Using the B3PW91/6-31G* level of theory and without imposing any symmetry constraint to the molecule, we get a bond length of 1.394 Å and the force constants for all bonds vary between 1212.5 and 14.0 kcal/mol Å², a variation of only ~0.1%, which is mainly due to truncation errors. For the six C—C—C bonds we get a force constant that vary between 200.9 and 201.0 kcal/mol deg², a deviation of only ~0.05%. Similar deviations are obtained for the CH bonds and CCH angles.

4. Conclusions

We have determined a very precise force field for alanine polypeptides that shows very little deviation for each type of bond. The relevance of the new force fields is that it allows performing simulations where vibrational modes are relevant; for example, cases where we need to insert signals into molecules and several other applications [14–18]. Cases where the vibrational modes are not of relevance; they are not affected by this new force field. The determination of parameters is unambiguous and from first principles; and it does not make use of any fitting procedure. The method can be extended to any system for which second derivatives are available.

Acknowledgements

We acknowledge financial support from the US Army Research Office and the US Defense Threat Reduction Agency (DTRA).

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